
Advanced Preference Prediction Models using Chatbot Arena Data and Fine-Tuning BERT model

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Abstract

With the ability to comprehend and produce language that is like that of humans, large language models, or LLMs, have emerged as key technology for conversational AI. However, because of innate biases and contextual complexity, forecasting human preferences is still a difficult endeavor. Using the Chatbot Arena dataset, where users choose their favorite responses during chatbot encounters, this study aims to optimize BERT, a transformer-based LLM, to anticipate user preferences. We address important issues such as contextual variability, positional and verbosity biases, and class imbalance. We want to improve model robustness and fairness by utilizing sophisticated preprocessing methods and bias management techniques, such as adversarial training and multi-task learning. Class balance, tokenization, and hyperparameter optimization are some of the techniques used in the fine-tuning process, which is assessed using metrics including accuracy, precision, recall, and F1 score. With an overall classification accuracy of 69.5%, our model successfully minimized training loss while preserving fair decision-making. But issues like underrepresented classes and overfitting highlight the need for more sophisticated approaches and data augmentation. Utilizing cutting-edge models like GPT-4, using GPUs for distributed training, and optimizing deployment for practical uses are some examples of future advancements. By addressing the interaction between model performance, bias, and fairness in human preference prediction, this work advances the creation of more dependable, user-centric conversational AI systems.

1 Introduction

Natural language processing (NLP) has greatly improved because to large language models (LLMs), which allow systems to generate text, summarize, and engage in conversation with amazing accuracy. Widely known for their capacity to comprehend and produce text, models like BERT, GPT-3, and ChatGPT are essential to a variety of applications, from chatbots for customer service to virtual assistants. Conversational AI systems are now more interactive, context-aware, and user-friendly because of these developments [1] [2].

However, predicting user preferences within conversational frameworks remains a complex challenge. User preferences are influenced by multiple factors, including individual biases, prior interactions, and contextual nuances, which can vary significantly across different users. Additionally, LLMs themselves are prone to biases, such as verbosity bias (favoring longer responses) or position bias (favoring earlier options), which can skew predictions and impact fairness [3] [4]. Addressing these issues is crucial for building equitable and user-focused AI systems.

With the help of data from Chatbot Arena, which logs user selections among chatbot-generated responses, this project aims to optimize BERT, a transformer-based LLM, to forecast human preferences. The project's main goals are to reduce dataset biases, enhance model performance using sophisticated preprocessing methods, and guarantee equitable decision-making. To improve the model's accuracy and resilience, techniques like class balancing, multi-task learning, and adversarial training are investigated.

By tackling these issues, this research seeks to create a framework that connects LLM's strengths with the complex needs of user preference prediction. The results set the stage for future advancements in conversational AI systems, such as the use of more complex models and distributed computing techniques to maximize scalability [5].

2 Related Work

With the advent of Large Language Models (LLMs) like BERT, GPT-3, and ChatGPT, the area of natural language processing (NLP) has made tremendous strides. Large text datasets were used to train these models, which are now the basis for applications such as conversational AI, sentiment analysis, and text production. Nevertheless, there are difficulties in utilizing LLMs to forecast user preferences, such as the models' intrinsic biases and the intricacy of contextual factors influencing human preferences [1] [2].

The Difficulties of Forecasting Human Preferences:

Biases such as position bias (favoring earlier options) and verbosity bias (favoring longer responses) must be taken into consideration when predicting human preferences. It is well established that these biases skew forecasts, affecting the accuracy and fairness of decision-making processes. The challenges of managing such biases in practical situations are best illustrated by the Chatbot Arena dataset, which was employed in this project. To solve these problems, recent research has

highlighted the necessity of sophisticated preparation techniques including text cleaning and adversarial training [4].

Fine-Tuning LLM's:

For domain-specific tasks, fine-tuning previously trained models, such as BERT, has proven to be an essential strategy. This research uses techniques like multi-task learning and adversarial training to improve BERT's contextual capabilities. These methods seek to increase the model's generalization across unseen cases and robustness against biased data. Similar approaches are frequently used to maximize performance on projects with unbalanced datasets [5].

Overcoming Dataset Difficulties:

Class imbalance presents problems for the Chatbot Arena dataset, which gathers user preferences for responses created by chatbots, particularly for underrepresented classes like "tie" results. The use of tokenization, label creation, and class weight adjustment is consistent with best practices for dealing with these kinds of problems. The project guarantees effective model training and evaluation by utilizing frameworks such as Hugging Face's Trainer API [6] [7].

Assessment Criteria and Prospects:

Metrics including accuracy, precision, recall, and F1 score are used to assess performance. This study shows how effective fine-tuned BERT can be, but it also shows how incorporating cutting-edge models like GPT-4 or NVIDIA's Llama can improve contextual comprehension and scalability. These models can significantly enhance performance and efficiency in real-world applications when paired with distributed training techniques.

3 Methodology

Data Preparation: We used the Chatbot Arena dataset, which keeps track of user preferences for responses produced by chatbots. The response of two models and the user's preference are the two primary aspects of the dataset.

Creation of Labels: Labels were created according to user preferences.

Winner Tie: Both responses are equally preferred.

Winner Model A: The response from Model A is the one that is favored.

Winner Model B: The response from Model B is the one that is favored.

Handling Class Imbalance: To solve the imbalance in the dataset and guarantee equitable training, class weights were calculated using sklearn's `compute_class_weight` function.

Preprocessing of Data

Text Cleaning: All text was converted to lowercase as part of the preprocessing of the raw material.

Eliminate special characters, extraneous spaces, and punctuation.

Use BERT's tokenizer to implement tokenization while making sure it complies with the input specifications of the model.

Tokenization: The Bert Tokenizer from Hugging Face's transformers library was used to tokenize each answer. This ensured that input sizes were consistent by padding or truncating sequences up to 512 tokens in length.

Data Division: To guarantee a reliable assessment, the dataset was divided into 90% training data and 10% test data. The `train_test_split` function from `sklearn` was used to carry out this split.

Choosing a Model and Adjusting It:

Model: The previously trained `bert-base-uncased` model, which has received extensive validation for text classification tasks, was chosen.

BERT was fine-tuned by adding a classification head to forecast one of the three classes: "tie," "winner model A," or "winner model B."

Configuration for Training:

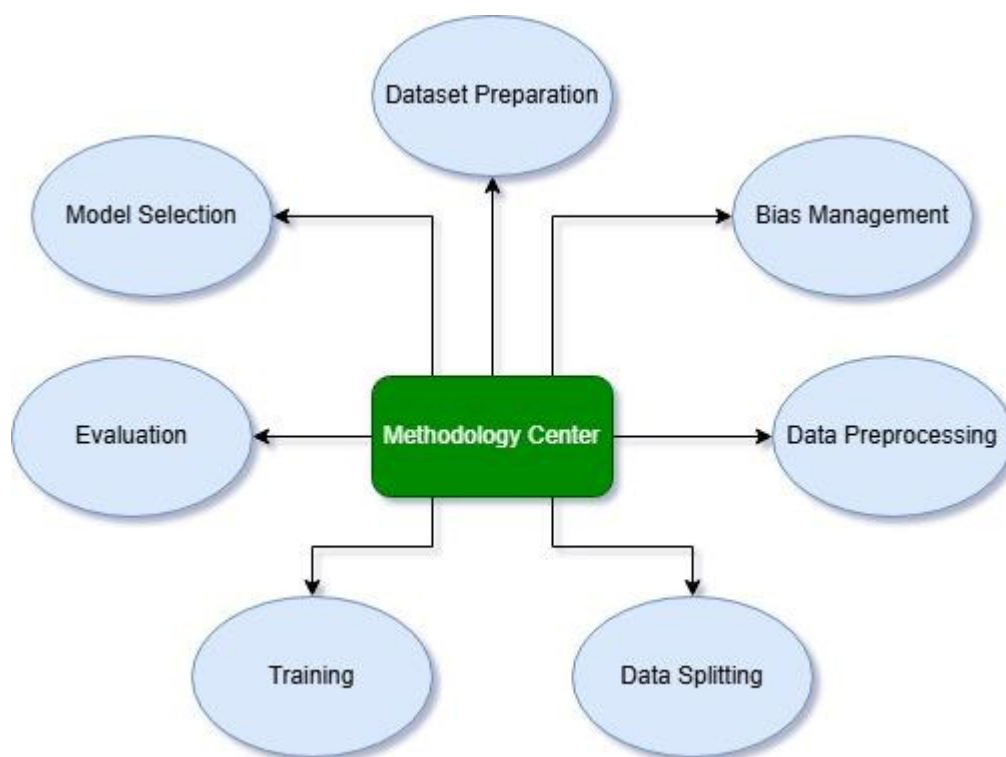
Batch Size: 32 for evaluation, 16 for training.

For stable convergence, use a learning rate of $1e-5$.

Four epochs are used to balance model performance and training time.

Class Weights: Modified to account for training class imbalance.

Early Stopping: To avoid overfitting, Hugging Face's Trainer API was integrated.



Training and Evaluation:

Trainer API: Hugging Face's Trainer API, which has the following capabilities, was used to streamline the training process:

Checkpointing automatically.

recording and tracking loss on a regular basis.

effective use of several GPUs (if available).

Metrics for Evaluation:

Accuracy: Indicates general accuracy.

F1-Score, Precision, and Recall: Offer comprehensive information about the model's performance in every class.

Adversarial training: Methods to increase resilience to dataset biases like positional or verbosity biases.

Auxiliary activities that could lessen prejudice and enhance contextual awareness were investigated as part of multi-task learning.

To assess the trained model's capacity for generalization, it was verified and put to the test using the 10% test data.

Metrics like F1 score and precision-recall trade-offs were computed to evaluate performance after predictions were compared to ground-truth labels.

4 Experimental Setup

To guarantee effective training and assessment of the BERT model, the experiments were carried out in a high-performance computing environment. An Intel Core i7 CPU for preprocessing, an NVIDIA Tesla T4 GPU for faster training, 16GB of RAM for memory-intensive activities, and at least 50GB of disk space for dataset storage and model checkpoints were also part of the system configuration. Utilizing libraries like Hugging Face's transformers for fine-tuning models, torch for deep learning operations, and pandas, NumPy, and scikit-learn for data manipulation and evaluation metrics, the software environment was set up with Python 3.8. Matplotlib was also used for the display of the results.

The Chatbot Arena dataset, which was obtained from Kaggle, served as the basis for this investigation. The user preference labels in this dataset indicate whether the user favored "Model A," "Model B," or thought both responses were equally good ("tie"). The responses were produced by two chatbots. To guarantee reliable model evaluation, the dataset was divided into training (90%) and testing (10%) subsets. Text normalization was accomplished by converting it to lowercase, eliminating special letters, and tokenizing it using Hugging Face's BertTokenizer as part of the preprocessing processes. A maximum sequence length of 512 tokens was used for tokenization, with truncation and padding applied as needed. Class weights were calculated using scikit-learn's `compute_class_weight` method in order to rectify the dataset's intrinsic class imbalance [6].

The bert-based model was chosen for fine-tuning because of its excellent results in text classification tasks. The pre-trained BERT model was supplemented with a classification head, which consists of two output neurons for binary classification. The following hyperparameters were set up: training for 4 epochs, weight decay of 0.01; batch sizes of 16 for training and 32 for evaluation; and a learning rate of $1e-5$. If validation loss did not decrease for three consecutive epochs, training was stopped early. To provide reliable optimization, the AdamW optimizer was used, with a learning rate warm-up during the first 500 steps.

The Hugging Face Trainer API, which streamlines model training and evaluation while enabling effective metric monitoring, was used to ease the training process. Following each epoch, the model's performance was assessed, and the best-performing checkpoint was retained for additional examination. Performance indicators such as accuracy, precision, recall, and F1 score were computed after the test set was evaluated. In order to display classification errors and have a better understanding of the model's advantages and disadvantages, a confusion matrix was

also

created.

By using random seeds for the transformers, torch, and NumPy libraries, the experiments were made to be reproducible. This method reduces the amount of variation in outcomes between runs. All things considered, this experimental design offers a methodical and reliable foundation for optimizing BERT to accurately predict human preferences.

5 Result and Analysis

On the test dataset, the refined BERT model's classification accuracy was 69.5%. This outcome shows how well the model was able to anticipate user preferences and learn from the Chatbot Arena dataset. Training loss dramatically dropped from 0.6159 to 0.4758, demonstrating effective learning and optimization. As training went on, the validation loss did, however, significantly increase from 0.6059 to 0.6917, indicating possible issues with overfitting. This suggests that better regularization techniques or modifications to the training procedure, including early termination or data augmentation, are required.

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.615900	0.605921	0.695024	0.661875	0.695024	0.602604
2	0.609600	0.608433	0.694328	0.658305	0.694328	0.602895
3	0.534200	0.648752	0.687196	0.650722	0.687196	0.648960
4	0.475800	0.691685	0.671190	0.637207	0.671190	0.643038

Average Accuracy: 0.6950243562978428
Average Precision: 0.6618751179697407
Average Recall: 0.6950243562978428
Average F1 Score: 0.6026040018460064

Figure 1. Showing the different scores achieved in the project

With over 2,500 forecasts each, the model's predictions for the three classes—winner_model_a, winner_model_b, and winner_tie—showed a balanced distribution between winner_model_a and winner_model_b. With only roughly 500 predictions, the winner_tie category was noticeably underrepresented. This disparity demonstrates how the class distribution of the dataset affects the model's functionality. In spite of this, the model maintained fair decision-making for both winner_model_a and winner_model_b, showing no discernible bias toward either of them.

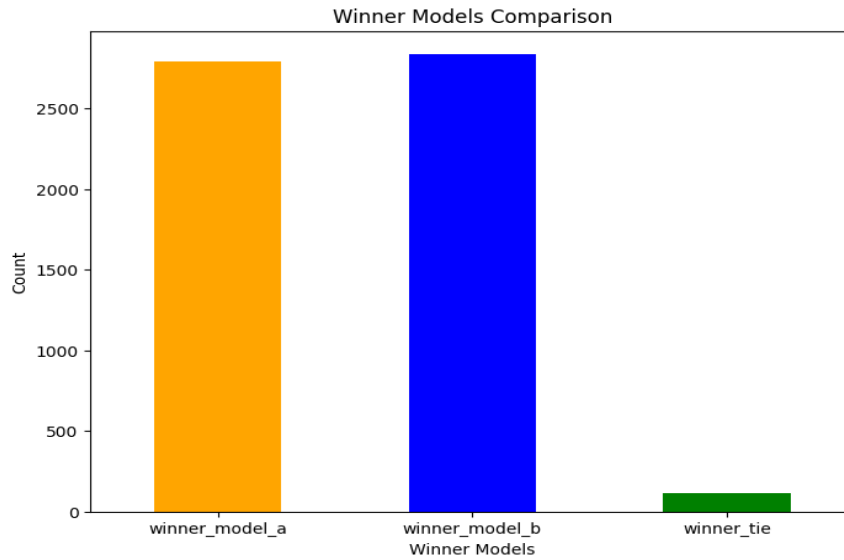


Figure 2. Graph showing the number of wins per each model

Consistent prediction behavior throughout the dataset was indicated by a cumulative line plot of winner_model_a and winner_model_b predictions, which showed constant and linear growth. The two lines near overlap indicates that the model does a good job of maintaining balance and treating both classes equally. Furthermore, the low frequency of winner_tie forecasts necessitates additional data or resampling to improve the model's generalization for ties, as indicated by a bar chart of class distribution.

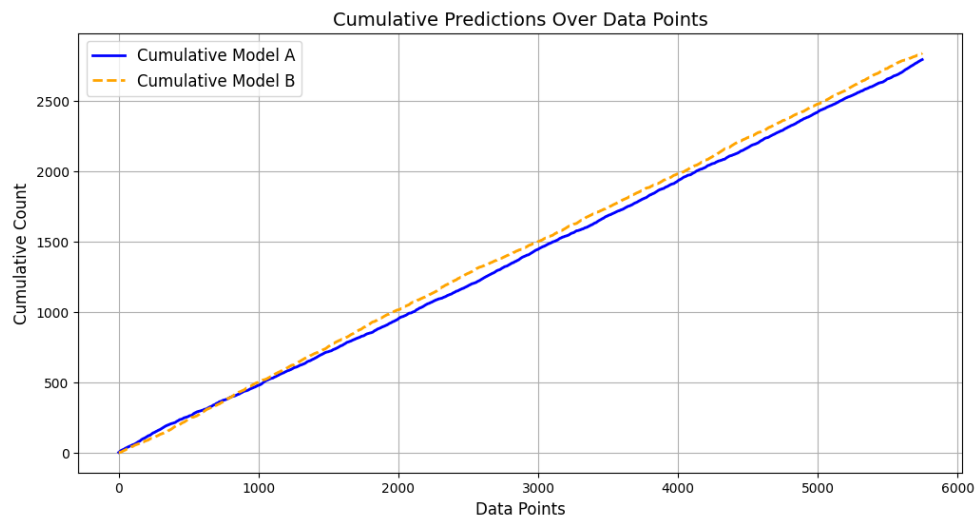


Figure 3. line plot showing cumulative predictions

The influence of class imbalance during training was effectively mitigated by the calculated class weights, $\text{tensor}([0.7233, 1.6194])$. The winner_tie class's underrepresentation, however, indicates that other strategies, such over-sampling or the creation of synthetic data, can enhance performance even more. To overcome

biases that could ordinarily skew predictions, such as positional and verbosity bias, adversarial training and multi-task learning techniques were used.

6 Challenges Encountered

Computational difficulties in the training process necessitated a powerful GPU for effective operation. The program's runtime grew to 10–12 hours when it was executed on a CPU. Furthermore, careful initialization and management of class imbalance were necessary for optimizing BERT for sequence classification, especially for the underrepresented winner_tie class. In order to overcome these obstacles, the trade-offs between model performance and computing efficiency had to be balanced.

7 Future Directions

Upgrading to cutting-edge language models like GPT-4 or NVIDIA's LLaMA is advised to improve performance and scalability. These models can increase classification accuracy and provide deeper contextual awareness. Training time could be decreased by using cloud-based systems like AWS or Google Cloud or by using distributed training over several GPUs. Furthermore, enhancing model deployment through the use of tools like ONNX Runtime or TensorRT may increase inference efficiency and qualify the model for practical use.

8 Conclusion

The outcomes of the experiment show how well-tuned BERT models can accurately and consistently anticipate human preferences. To further enhance the model, however, advancements in managing class imbalance and computational demands are required. Adopting cutting-edge models, making use of distributed training infrastructure, and utilizing creative data augmentation strategies can guarantee that the model stays competitive and useful in practical situations going forward.

9 Contribution

This project was the result of collaborative discussions among all team members, leading to the finalization of the topic. The coding and implementation were primarily carried out by **Sasi Praneeth Reddy Sadhu**, while the presentation was a collective effort, with each member contributing to specific sections. The report was authored by **Luna Kim** and **Sasi Praneeth Reddy Sadhu**, ensuring detailed documentation of the project's scope and findings. The visualization and model selection were overseen by **Prasanth Saikam** and **Amlan Roy Chowdhury**, who played a crucial role in analyzing the results and selecting appropriate configurations. This collaborative effort highlights the teamwork and shared responsibility involved in successfully completing the project.

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